Q: If you had to design a 24-bit ADC with 100 kHz bandwidth, what would be the hardest problem?

A AI Model: Gemma4

Q: If you had to design a 24-bit ADC with 100 kHz bandwidth, what would be the hardest problem?
Time: (1m 30.43s)
A:
The hardest problem is maintaining both the extremely stringent 24-bit linearity requirement ($\mathrm{INL/DNL} < 0.5\mathrm{LSB}$) and the necessary Signal-to-Noise Ratio ($\mathrm{SNR} > 148\mathrm{dB}$ for $\mathrm{ENOB} = 24$) across a $100\mathrm{kHz}$ bandwidth without excessive power consumption. While high speed typically stresses comparator delay or clock jitter, at this combination of specifications, the challenge shifts to **managing input-referred noise integration and component matching uniformity**; any residual non-ideality is amplified into an unacceptable error floor. Specifically, achieving $\mathrm{ENOB} = 24$ mandates that the total integrated thermal noise (σ_{noise}) from the front-end transimpedance stage or sample-and-hold network must be below $\approx 0.5\mu\mathrm{V}_{\mathrm{rms}}$, which requires meticulous circuit design—especially concerning capacitor matching (for DACs, e.g., in SAR/Pipeline architectures) and op-amp gain bandwidth product (GBW) consistency across all parallel paths to ensure that the non-linearity introduced by finite OpAmp slew rate or settling time does not manifest as a degradation of the lower bits' Effective Number of Bits (ENOB).).

B AI Model: Llama3.1

Q: If you had to design a 24-bit ADC with 100 kHz bandwidth, what would be the hardest problem?
Time: (1m 7.4s)
A:
The hardest problem in designing a 24-bit ADC with 100 kHz bandwidth is achieving high resolution while maintaining sufficient signal-to-noise ratio (SNR) and linearity. In particular,

the challenge lies in minimizing the quantization noise power, which increases exponentially with increasing resolution. For a 24-bit ADC, the maximum quantization error is ±2^23 LSBs, corresponding to an SNR of approximately 147 dB. To achieve this, the ADC must have an input signal-to-noise ratio (SNR) of at least 150-160 dB, which is difficult to obtain with practical circuit techniques.

Let's consider the thermal noise power spectral density (PSD) of a typical op-amp used in an SAR ADC: Nt(f) = 4kT/Rin ≈ -137 dBFS/Hz. To achieve an SNR of 150 dB at 100 kHz bandwidth, the input signal amplitude would need to be approximately 10^15 V or 3000 V peak-to-peak, which is impractically large. Even with a well-designed ADC architecture and circuit techniques, such as correlated double sampling and chopper stabilization, it's challenging to achieve an SNR of more than 140 dB at 100 kHz bandwidth.

Furthermore, the linearity requirements for a 24-bit ADC are extremely stringent. The differential non-linearity (DNL) must be better than ±0.25 LSB, while the integral non-linearity (INL) should be within ±1 LSB. Achieving these levels of linearity requires careful design and implementation of the ADC's digital-to-analog converter (DAC), comparator, and analog circuitry.

In summary, the hardest problem in designing a 24-bit ADC with 100 kHz bandwidth is striking an optimal balance between high resolution, sufficient SNR, and practical implementation constraints..

C AI Model: Qwen3.5

Q: If you had to design a 24-bit ADC with 100 kHz bandwidth, what would be the hardest problem?
Time: (1h 2m 19.27s)
A:
For a 24-bit ADC with this low-to-moderate bandwidth, the fundamental design bottleneck is maintaining **Integral Non-Linearity (INL) below the noise floor** while simultaneously satisfying thermal stability requirements. In successive-approximation architectures typical for such specifications, capacitor matching precision must exceed $\sim$ppm levels to keep DNL/INL errors within $\pm 0.5\ \mathrm{LSB}$ ($V_{LSB} = V_{ref}/2^{N-1}$), where any systematic mismatch degrades SINAD disproportionately (harmonic distortion scales with $C_{mismatch}$). Consequently, the hardest engineering challenge is balancing these linearity demands against **power consumption** and area—specifically, minimizing thermal noise (kT/C) from switch arrays

while integrating high-order digital reference calibration or trimming circuitry without exceeding budget constraints..

D AI Model: Ministral-3

Q: If you had to design a 24-bit ADC with 100 kHz bandwidth, what would be the hardest problem?
Time: (2m 12.25s)
A:
The **hardest challenge** in designing a **24-bit ADC with a 100 kHz bandwidth** is achieving **low noise and distortion while maintaining high dynamic range (DR) and signal-to-noise-and-distortion ratio (SNDR)**—a conflict exacerbated by the **Nyquist rate constraint** (sampling at ⩾200 kS/s for 100 kHz bandwidth). The primary obstacles are:

1. **Thermal Noise Dominance**: At 24-bit resolution, the **quantization noise floor** is ~−176 dBFS (for full-scale input), but **thermal noise** in the front-end (e.g., input buffer, sampling capacitor) and **kTC noise** from the sampling switch can exceed this if not carefully suppressed. For example, a 24-bit ADC requires **<10 nV RMS thermal noise** at the input (assuming 1 V full-scale), which demands ultra-low-noise op-amps (e.g., GBW >1 GHz, CMRR >130 dB) and high-quality capacitors (low ESR/ESL).

2. **Nonlinearity and INL/DNL Errors**: Achieving **<±0.5 LSB INL** (≈4 µV for 1 V full-scale) requires **precise capacitor matching** (±0.003% for SAR ADCs or ±0.01% for pipeline stages) and **low comparator offset** (<1 mV), which is challenging at high resolution due to process variations, charge injection, and clock feedthrough.

3. **Clock Jitter**: Sampling jitter (e.g., 5 ps RMS) degrades SNDR by **−20 log(ω・τ)** (where ω = 2π・100 kHz), introducing **>−90 dBFS noise** at the Nyquist frequency—far above the quantization floor. Low-jitter oscillators (e.g., <1 ps RMS) and careful layout shielding are mandatory.

4. **Power Efficiency vs. Performance**: A **pipeline ADC** could achieve 200 kS/s with moderate power (~50 mW), but **multistage gain errors** and **metastability in comparators** degrade performance. A **ΔΣ ADC** (e.g., 3rd-order oversampling at ×16) could relax the sampling rate to ~12.5 kS/s while improving noise shaping, but it requires **high loop bandwidth** (>10 MHz) and **low noise amplifiers**, increasing power (~100 mW). A **hybrid SAR/pipeline approach** (e.g., 18-bit SAR front-end + pipeline) might balance latency and efficiency but adds complexity.

5. **Reference Stability**: A **24-bit ADC demands <±1 ppm reference stability** over temperature/voltage, requiring ultra-stable bandgap references or external precision sources (e.g., ±0.5 mV for 1 V full-scale).

Most critical trade-off: The **Nyquist rate requirement forces high-speed sampling**, which increases **kTC noise, comparator metastability risk, and power consumption**. Without oversampling (e.g., ΔΣ) or a relaxed bandwidth specification, the design becomes impractical with current technology. For highest performance, a **ΔΣ ADC with 3rd-order noise shaping** is the most robust choice, though it trades latency for improved SNDR..
